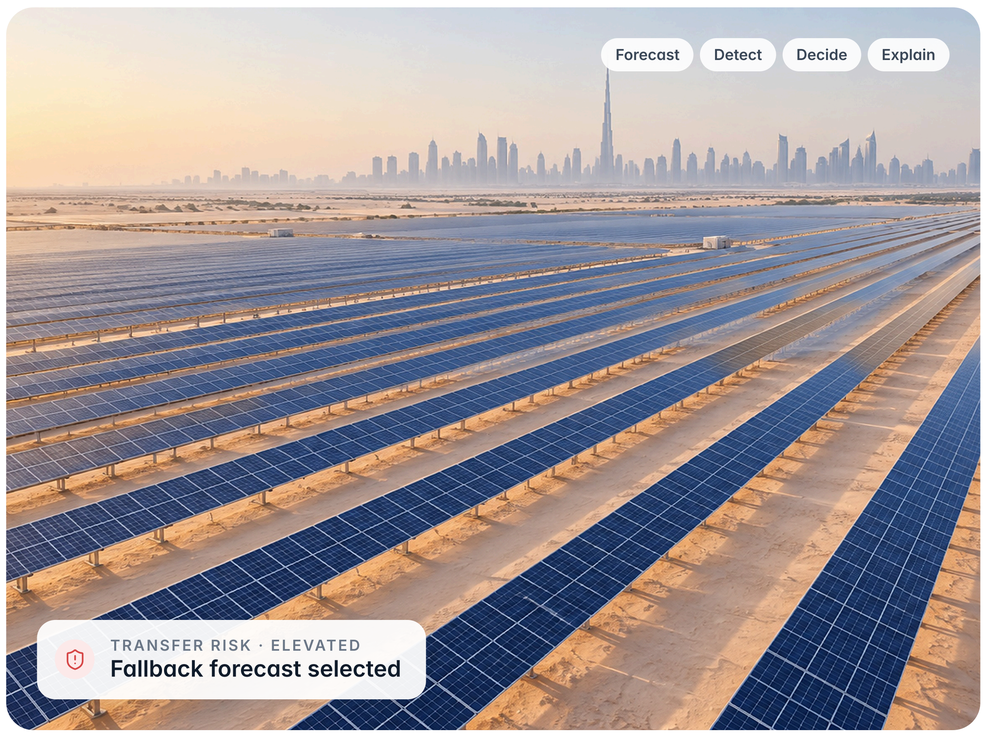
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RESEARCH POSTER · SMART ENERGY SYSTEMS

SolarTwin AI

A trustworthy agentic digital twin for adaptive solar energy management



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01 ABSTRACT

Not just what a new solar plant will generate, but when to trust the forecast, and how to act on it safely.

New plants have little generation history, so operators reuse models trained on other sites. Our accepted NEECON 2026 paper shows that this transfer can succeed, and can also fail. **SolarTwin AI** is a software digital twin that monitors transfer risk, compares forecasts, plans battery and load actions within hard limits, and explains every decision.

02 PROBLEM & OBJECTIVES

A transferred forecast can be confidently wrong, and nothing in a standard pipeline flags it.

01 **Detect** when a transferred forecast should not be trusted.

02 **Decide** a safe battery and flexible-load action within constraints.

03 **Explain** and log each decision with measured evidence.

03 METHODOLOGY

Data & scenarios
Plant telemetry, weather inputs, controlled shift events

Forecast engine
CDAN + LSTM + Clear-Sky Index · Transformer · Persistence

Trust monitor
Feature shift · model disagreement · risk flag

Energy optimizer
Battery and flexible load under hard constraints

Explain & audit
Decision trace; every number computed in code

INTERACTIVE DIGITAL TWIN · CLOSED LOOP

NEW A trust layer between forecasting and control. Agents call real models and solvers; the LLM explains, but never invents numbers or overrides constraints.

04 RESULTS & ANALYSIS

ACCEPTED PAPER · INTRA-REGIONAL

0.8325 R^2

Plant 1 → Plant 2 with zero target labels, using CDAN + LSTM + Clear-Sky Index under capacity mismatch.

ACCEPTED PAPER · CROSS-REGIONAL

Negative transfer

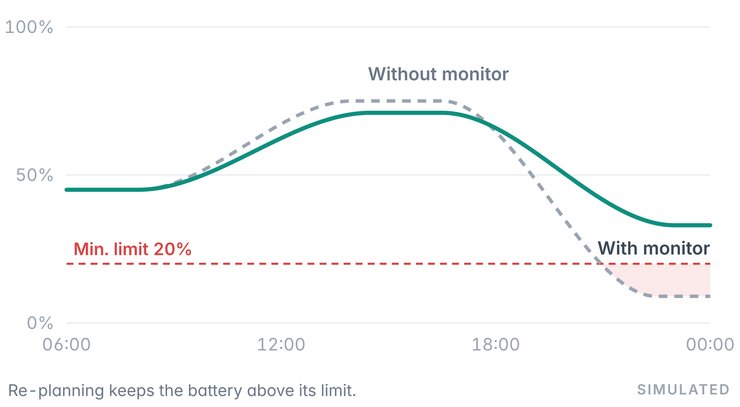
India → Australia: adaptation hurt accuracy, while an unadapted Transformer kept positive skill.

IMPLICATION

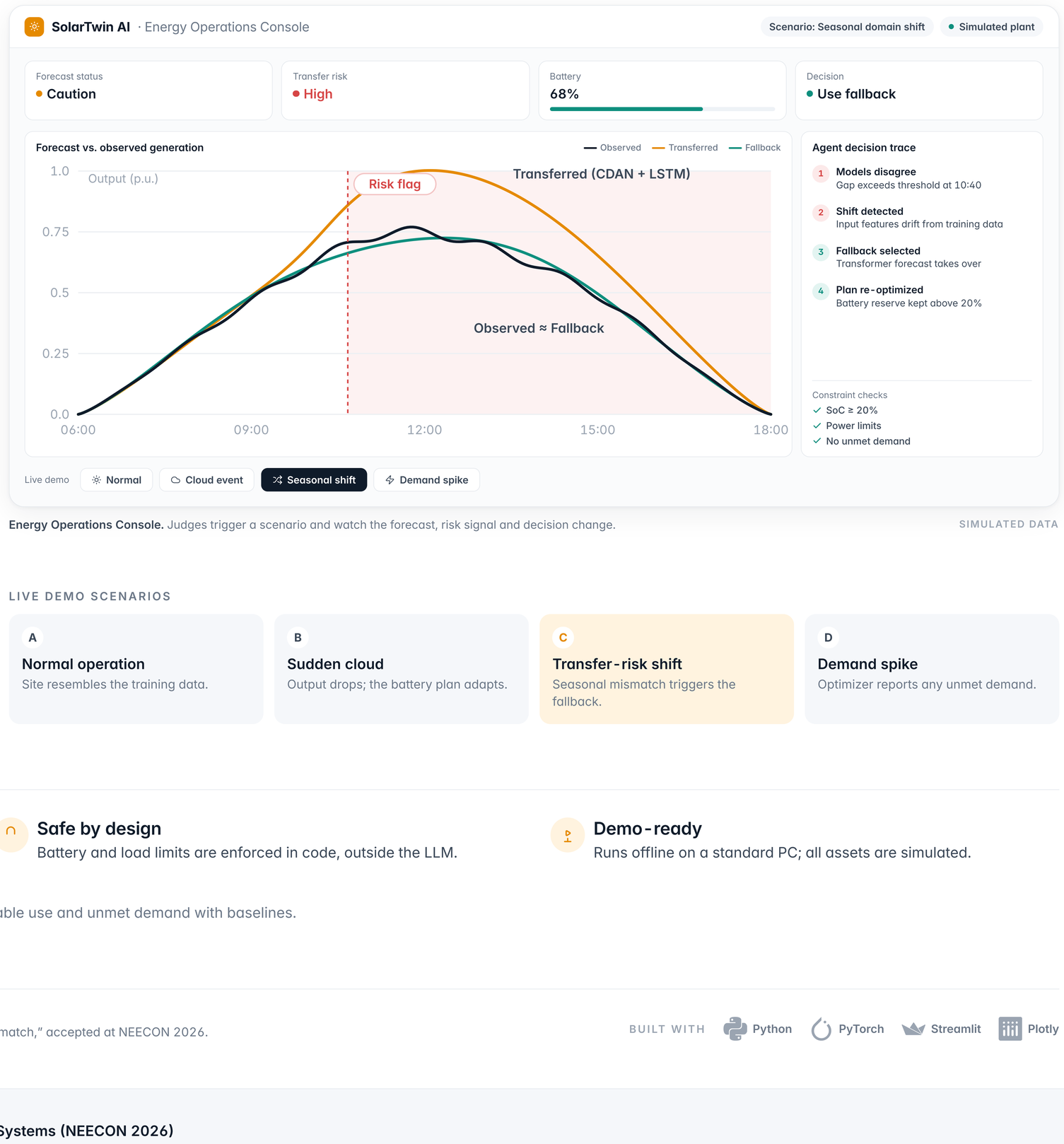
Monitor, don't assume

Transfer quality depends on the site. SolarTwin AI checks it before a forecast drives a decision.

BATTERY STATE OF CHARGE



Re-planning keeps the battery above its limit. SIMULATED



Energy Operations Console. Judges trigger a scenario and watch the forecast, risk signal and decision change. SIMULATED DATA

LIVE DEMO SCENARIOS

A — Normal operation
Site resembles the training data.

B — Sudden cloud
Output drops; the battery plan adapts.

C — Transfer-risk shift
Seasonal mismatch triggers the fallback.

D — Demand spike
Optimizer reports any unmet demand.

05 CONCLUSIONS

Trust is visible
Risk flags, model disagreement and fallbacks are shown, not hidden.

Safe by design
Battery and load limits are enforced in code, outside the LLM.

Demo-ready
Runs offline on a standard PC; all assets are simulated.

Next: report detector precision, recall and false alarms, and compare cost, renewable use and unmet demand with baselines.

06 REFERENCES & CONTACT

[1] Belay et al., "Learning When to Transfer: Robust Solar Power Forecasting Under Capacity Mismatch," accepted at NEECON 2026.

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